

Users > moshika > Desktop > lab >  family.pl

```
1  % ----- Facts -----
2
3  male(john).
4  male(michael).
5  male(david).
6  male(sam).
7  male(rohan).
8
9  female(sarah).
10 female(linda).
11 female(emma).
12 female(sophia).
13 female(rita).
14
15 parent(michael, john).
16 parent(michael, sarah).
17 parent(emma, john).
18 parent(emma, sarah).
19
20 parent(david, michael).
21 parent(sophia, michael).
22
23 parent(sam, linda).
24 parent(sam, david).
25
26 parent(rohan, david).
27 parent(rohan, rita).
28
```

```
29  % ----- Rules -----
30
31  father(Child, Father) :-
32      parent(Child, Father),
33      male(Father).
34
35  mother(Child, Mother) :-
36      parent(Child, Mother),
37      female(Mother).
38
39  sibling(A, B) :-
40      parent(A, P),
41      parent(B, P),
42      A \= B.
43
44  brother(A, B) :-
45      sibling(A, B),
46      male(A).
47
48  sister(A, B) :-
49      sibling(A, B),
50      female(A).
51
52  grandparent(C, G) :-
53      parent(C, P),
54      parent(P, G).
55
56  grandfather(C, Gf) :-
57      grandparent(C, Gf),
58      male(Gf).
59
60  grandmother(C, Gm) :-
61      grandparent(C, Gm),
62      female(Gm).
```

```

63
64 uncle(C, U) :-
65     parent(C, P),
66     sibling(P, U),
67     male(U).
68
69 aunt(C, A) :-
70     parent(C, P),
71     sibling(P, A),
72     female(A).
73
74 cousin(A, B) :-
75     parent(A, P1),
76     parent(B, P2),
77     sibling(P1, P2).
78
79 % ----- User Input / Interactive -----
80
81 start :-
82     write('Enter a person name: '),
83     read(Name),
84
85     ( father(Name, F), write('Father: '), write(F), nl ; true ),
86     ( mother(Name, M), write('Mother: '), write(M), nl ; true ),
87     ( brother(Name, B), write('Brother: '), write(B), nl ; true ),
88     ( sister(Name, S), write('Sister: '), write(S), nl ; true ),
89     ( grandfather(Name, GF), write('Grandfather: '), write(GF), nl ; true ),
90     ( grandmother(Name, GM), write('Grandmother: '), write(GM), nl ; true ),
91     ( uncle(Name, U), write('Uncle: '), write(U), nl ; true ),
92     ( aunt(Name, A), write('Aunt: '), write(A), nl ; true ),
93     ( cousin(Name, C), write('Cousin: '), write(C), nl ; true ),
94
95     nl, write('Query complete. '), nl.

```

```
[moshika@Moshikas-MacBook-Air ~ % cd /Users/moshika/Desktop/lab  
[moshika@Moshikas-MacBook-Air lab % swipl family.pl  
Welcome to SWI-Prolog (threaded, 64 bits, version 9.2.9)  
SWI-Prolog comes with ABSOLUTELY NO WARRANTY. This is free software.  
Please run ?- license. for legal details.
```

```
For online help and background, visit https://www.swi-prolog.org  
For built-in help, use ?- help(Topic). or ?- apropos(Word).
```

```
[?- start.  
[Enter a person name: rohan.  
Father: david  
Mother: rita  
Brother: sam  
Grandfather: michael  
Aunt: sophia
```

```
Query complete.  
true .
```

```
[?- halt.
```